

Due date: June 5, 2026

Late submission: 20% per day.

Teams: You can do the project in teams of 2. Teams must submit only 1 copy of the project.

Optimizing Hospital Locations Using Search and Optimization Algorithms

Project Objective:

Your task is to design and implement two **optimization algorithms** that decide:

- **Where to build hospitals,** and
 - **How many hospitals to build,**
- in order to balance **patient accessibility** and **construction cost**.

Problem Overview

You are given:

- A set of population locations (people who need access to hospitals)
 - A set of possible locations where hospitals can be built
- You must choose a subset of these candidate locations to build hospitals. Each person will go to their nearest hospital.

The challenge is:

- Adding more hospitals reduces travel distance
 - But building hospitals is expensive
- Your algorithm must find a good balance between these two competing objectives.

Input to Your Algorithm

Your program should accept the following:

1. Population Points: A list of coordinates: $P = [(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)]$
2. Population Weights: Each point may have a weight (number of people): $W = [w_1, w_2, \dots, w_n]$
3. Candidate Hospital Locations: A list of possible sites: $C = [(c_{1x}, c_{1y}), (c_{2x}, c_{2y}), \dots, (c_{mx}, c_{my})]$
4. Hospital Cost Parameter (λ): A positive number $\lambda > 0$, This controls how expensive it is to build hospitals.

A solution is a subset of candidate locations: $H \subseteq C$

Search Method

- Implement at least two of:
 1. Hill Climbing
 2. Simulated Annealing
 3. Genetic Algorithm
- Compare algorithms on:
 - Final cost
 - Runtime
 - Stability (variance across runs)
- Run experiments with different values of λ
- Analyze how λ affects:
 - Number of hospitals
 - Travel distances

Submission:

- A report of up to 8 pages that describes:
 - o Problem formulation (with cost function)
 - o Experimental setup
 - o Results (tables/plots)
 - o Effects of parameters tuning (at least one parameter per algorithm)
 - o Visualizations of your solutions
 - o ConclusionsDo not include code snippets in the report
- Source code: well-documented implementations of both algorithms with clear input/output format
- Demo: you will be asked to present a demo of your project. During the demo, you should expect questions about your code and you might be asked to apply some minor modification.

Suggested Baseline Experimental Setup:

1) Problem Instance

Population points

- Number of points: $n = 100$
- Coordinates: $(x, y) \in [0, 100] \times [0, 100]$
- Distribution: Uniform random

Population weights

- $w_i \in [1, 10]$ (integers, uniform)

This introduces moderate variability without dominating the distance term.

Candidate hospital locations

- Number of candidates: $m = 100$
- Coordinates: Same space $[0, 100] \times [0, 100]$
- Generated independently of population points

Distance metric

- Euclidean distance

Hospital cost parameter (λ)

Use a log-scale range:

- $\lambda \in \{1, 10, 50, 100\}$

2) Algorithm Baselines

A) Hill Climbing

- Initialization: Random solution with ~10–20% hospitals selected
- Max iterations: 500

B) Simulated Annealing

- Initial temperature: $T_0 = 1000$
- Cooling rate: $\alpha = 0.95$

C) Genetic Algorithm

Parameters:

- Population size: 200
- Generations: 100
- Mutation rate: 0.05

Honor Policy: All are required to adhere to the university honor policy and violations will be dealt with according to university regulations.